Dhungana *Energy Informatics* (2025) 8:2
https://doi.org/10.1186/s42162-024-00459-4

Energy Informatics

REVIEW

Open Access

A machine learning approach for wind turbine power forecasting for maintenance planning

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Abstract
Integrating power forecasting with wind turbine maintenance planning enables an innovative, data-driven approach that maximizes energy output by predicting periods low wind production and aligning them with maintenance schedules, improving operational efficiency. Recently, many countries have met renewable energy targets, primarily using wind and solar, to promote sustainable growth and reduce emissions. Forecasting wind turbine power production is crucial for maintaining a stable and reliable power grid. As renewable energy integration increases, precise electricity demand forecasting becomes essential at every power system level. This study presents and compares nine machine learning (ML) methods for forecasting, Interpretable ML, Explainable ML, and Blackbox model. The interpretable ML includes Linear Regression (LR), K-Nearest Neighbors (KNN), eXtreme Gradient Boosting (XGBoost), Random Forest (RF); the explainable ML consists of graphical Neural network (GNN); and the blackbox model includes Multi-layer Perceptron (MLP), Recurrent Neural Network (RNN), Gated Recurrent Unit (GRU), and Long Short-Term Memory (LSTM). These methods are applied to the EDP datasets using three causal variable types: including temporal information, metrological information, and power curtailment information. Computational results show that the GNN-based forecasting model outperforms other benchmark methods regarding power forecasting accuracy. However, when considering computational resources such as memory and processing time, the XGBoost model provides optimal results, offering faster processing and reduced memory usage. Furthermore, we present forecasting results for various time windows and horizons, ranging from 10 minutes to a day.

Keywords Machine learning, Deep learning, Energy forecasting, Condition monitoring, Wind turbine

Introduction

Wind turbine power generation is expected to grow significantly over the next decade [1]. This is essential for reducing energy costs, maintaining reliable electricity supply, and supporting the transition to a sustainable energy future. Integrating wind power into existing power grids presents various challenges for system operators, such as managing operating reserves, unit scheduling, transient stability, and transmission congestion [2]. Power forecasting is regarded as a vital tool for grid integration, aiding decision-making, reducing imbalances, and enhancing grid stability [3]. Accurate forecasts help balance supply and demand, ensuring grid stability and optimizing the integration of renewable energy for energy planning[4, 5].

Condition monitoring, decision-making, and maintenance planning are closely interconnected in asset management. Condition monitoring involves collecting real-time data on equipment health to identify potential issues [6]. This data informs decision-making, enabling timely actions to prevent failures. These decisions then guide maintenance planning, optimizing schedules, resources, and interventions to ensure reliability and cost-efficiency [7]. Effective decision-making relies on comprehensive insights derived from fault detection, identification, quantification, and prognosis to ensure accurate assessment and timely action [8]. Prognosis or forecasting, regardless of the term used, predicts future outcomes based on current data and trends. Forecasting supports planning in industrial maintenance by predicting future equipment degradation conditions, resource needs, and failures, enabling proactive interventions. Integrating power forecasting with maintenance planning optimizes operations by aligning maintenance schedules with low wind periods, maximizing energy output [2]. Together, they form a feedback loop that enhances operational performance and maximizes energy sales.

Certain limitations are evident in wind power forecasting. For instance, Some studies have primarily focused on a limited set of parameters, such as wind speed, as the key input for predictions [9]. While wind speed is undeniably a critical factor, it is not the only determinant of wind power generation. Other parameters, including wind direction, air density, temperature, turbine operational conditions, and curtailment data, also play significant roles in influencing wind power output. Some studies rely on only one or two evaluation metrics, such as RMSE or MAE, to assess forecasting performance, which can provide a limited understanding of the model's overall effectiveness [3]. Some employed disproportionate data splits, allocating over 98% of the data for training and less than 2% for testing [10] . Such practice may artificially inflate accuracy metrics while potentially overlooking the model's generalization and real-world applicability.

This study evaluates a broader range of ML models to address these gaps by incorporating all relevant temporal parameters, meteorological factors, and turbine curtailment data. This paper comprehensively explains the step-by-step forecasting procedure, including techniques for reconstructing missing values, creating time-based features to account for short-term and long-term weather patterns, outlier removal methods, and the implementation of machine learning models. We emphasize the significant contribution of analyzing and comparing nine different ML models for power forecasting. This comprehensive evaluation not only advances our understanding of how various algorithms perform under different conditions but also highlights the strengths and weaknesses of each model. It also ensures a more balanced data division of 80% for training and 20% for testing while considering a comprehensive set of deterministic metrics to provide a holistic assessment of model performance. By identifying the most accurate and reliable models, the study contributes to improving prediction accuracy, which is crucial for optimizing grid management and enhancing the integration of wind energy into the power system. This analysis is valuable for researchers and practitioners aiming to develop more effective forecasting tools in the renewable energy sector.

The remainder of the paper is structured as follows: Section 2 reviews relevant literature. Section 3 outlines the theoretical concepts of the wind power calculation method and the associated variables. Section 4 details the methodology for power forecasting, including step-by-step data preprocessing, feature engineering, and ML-based forecasting models. Section 5 describes the data, evaluation metrics, and benchmarks. Section 6 presents comprehensive results at each stage. Section 7 analyzes these results and discusses the findings of this research. Finally, Section 8 concludes the paper and offers an outlook on future implications and potential extensions.

Literature review

In recent years, considerable efforts have been made to develop more accurate forecasting models. Various ML models for wind power prediction are discussed, highlighting the parameters used, evaluation metrics, and proposed directions for future work, as elaborated in [2]. Data processing methods in wind energy forecasting were reviewed and categorized into seven groups in [11]. These methods were evaluated from different perspectives, and their trends and challenges were discussed. There is no consensus on the number of categories for classifying the forecasting horizon in wind energy [2, 12]. However, short-term, medium-term, and long-term forecasting are widely used. Short-term forecasts, which focus on hours or minutes, assist in adjusting system reserves and managing economic dispatch to stabilize the grid. Medium-term forecasts, spanning days to weeks, support equipment maintenance and troubleshooting. Long-term forecasts cover weeks to months and are utilized for wind farm planning and resource evaluation.

Forecasting methods can be categorized into physical, statistical, artificial intelligence, and hybrid approaches [10]. Physical methods rely on meteorological data, are less dependent on historical power data, and are sensitive to initial conditions. Statistical methods use historical data and various models like Markov chains and ARMA to make predictions, requiring substantial data but offering high accuracy. The research shows that ML-based models can accurately forecast wind and provide crucial information about potential wind use locations for power plants by predicting future wind speeds [4]. Deep learning based models apply complex relationships between inputs and outputs, improving prediction accuracy. Hybrid models combine multiple AI techniques to address limitations of individual methods, enhancing forecasting capabilities [13].

There are two primary approaches to forecasting: deterministic and probabilistic [12]. The deterministic forecasting method provides a single, specific prediction or outcome based on input data and the model [14]. Probabilistic forecasting provides a range of possible outcomes with associated probabilities, acknowledging uncertainty by offering a distribution of future states [5, 15]. It is valuable when uncertainty is high, risk needs to be quantified, or decisions depend on understanding various possible outcomes. Deterministic forecasting delivers precision and simplicity, while probabilistic forecasting offers a more comprehensive view by accounting for uncertainty. Forecasting can be either single-step or multi-step, depending on the step size [16]. Single-step forecasting predicts the next value based on the previous n observations, while multi-step forecasting predicts multiple future values by capturing complex trends. Although multi-step forecasting provides insights over a longer period, its accuracy tends to decrease as the prediction horizon extends. Wind power forecasting models can be categorized as parametric, which assumes specific data distributions for efficiency, and nonparametric, which makes no such assumptions and is robust for diverse data [1].

Forecasting models can be applied using either a direct or an indirect approach. The direct approach predicts power output directly, and they are simpler and quicker, while the indirect approach first forecasts wind speed with weather data and then convert it to power estimates using the turbine's power curve [17]. Weather data transformations for input into forecasting models were analyzed in [18] to enhance forecast accuracy. Methodologies for predicting power production using SCADA data, employing power curve parametric using XGBoost models to handle variability, and a Possibilistic Approach to address uncertainty was used in [19].The forecasting model using empirical mode decomposition, K-means clustering, and machine learning showed improved accuracy for short-term wind speed and power forecasting but was less effective over longer durations [13]. Table 1 provides a concise overview of ML-based approaches for forecasting.

Wind power and associated variables

Wind speed is the primary factor influencing wind turbine power forecasting, and wind power forecasting presents significant challenges due to the random, intermittent, and uncontrollable nature of wind speed. The amount of wind energy $P_{\text{Electrical}}$ that can be harnessed by wind turbines is represented by Eqs.(1), [19]. Where *v* represents wind speed, which determines the amount of energy available, with power output increasing with higher wind speeds until reaching the turbine's rated capacity. ρ represents air density; it influences the total energy available in the wind, with denser air providing more energy per unit volume. *A* represents the rotor disc area. C_{p} represent rotor power coefficient which signifies the wind energy utilization factor by the rotor based on tip speed radio (λ) and blade pitch angle (β) along with relative wind direction, how effectively the turbine captures optimum power generation. η represents the mechanical and electrical efficiency of wind turbine [16].

$$\begin{aligned} P_{\text{Electrical}} &= P_{\text{Rotor}} \times \eta \\ &= P_{\text{Wind}} \times C_p(\lambda, \beta) \times \eta \\ &= \frac{1}{2}\rho A v^3 \times C_p(\lambda, \beta) \times \eta \end{aligned} \tag{1}$$

$$\text{Air density } (\rho) = \rho_d + \rho_v \tag{2}$$

$$C_p(\lambda, \beta) = 0.5(\Gamma - 0.022\beta^2 - 5.6)e^{-0.17\Gamma} \tag{3}$$

Where

$$\begin{aligned} \rho_d =& \frac{P - P_v}{R_{\text{Specific, Dry air}} \times T_k}, \quad \rho_v = \frac{P_v}{R_{\text{Specific, Water vapour}} \times T_k}, \quad P_v = \frac{h \times P_{\text{sat}}}{100}, \\ P_{\text{sat}} =& 6.1078 \times 10^{\frac{7.5T}{T+237.3}}, \quad \lambda = \frac{\omega_{wtb} R}{v}, \quad \text{and} \quad \Gamma = \frac{R}{\lambda} \cdot \frac{3600}{1609} \end{aligned}$$

ρ_d represents density of dry air; ρ_v represents density of water vapour; P_v represents partial pressure of water vapour; $R_{Specific}$ represents specific gas constant; T_K represents temperature in Kelvin; *h* represents humidity; P_{sat} represents saturation water vapour pressure; ω_{wtb} is the angular velocity of the wind turbine; *R* represents the blade radius.

Meteorological data, such as wind speed and air density, both influenced by ambient temperature, humidity, and atmospheric pressure, vary throughout the day.

Table 1 Summary of literature review

Refe	Characteristics of forecasting models	Types		
		Direct/Indirect	**Deterministic/Probabilistic**	**Single-step/ Multi-step**
[3]	Gradient boosting method for short- and medium-term wind power prediction	Direct	Deterministic	Single-step
[5]	Dual-stage attention based LSTM network for load forecasting	Direct	Both	Both
[10]	A hybrid NN-based day-ahead forecasting model	Direct	Deterministic	Single-step
[14]	Explainable causal GNN network for multivariate forecasting	Direct	Deterministic	Multi-step
[15]	Probabilistic load forecasts using quantile regression averaging	Direct	Probabilistic	Multi-step
[16]	An LSTM network was employed to replicate Savitzky-Golay filter results, effectively mitigating overfitting caused by noise and improving generalization	Direct	Deterministic	Both
[17]	CNN-based numerical weather modeling for reliable wind power forecasting	Indirect	Deterministic	Single-step
[18]	Investigate transformations of weather data inputs for accurate forecasting.	Indirect	Deterministic	Multi-step
[20]	Various ML techniques, including LR , SVM, GPR, and ANN, are evaluated to identify the most efficient method for short-term load forecasting	Direct	Both	Single-step
[21]	Fuzzy-based approach for average monthly wind power forecasting	Direct	Deterministic	Single-step
[22]	Dual-meta pool model for small sample data,	Direct	Deterministic	Multi-step
[23]	Multi-source data fusion for wind turbine power forecasting	Direct	Deterministic	Single-step
[24]	Three deep learning based (RNN, LSTM, GRU) forecasting model	Direct	Deterministic	Single-step
[25]	Advanced ML techniques multi-task learning and generative critic networks for wind power forecasting	Direct	Probabilistic	Multi-step
[26]	Wind power production forecasting from weather research and forecasting (WRF) model using machine learning	Direct	Deterministic	Single-step

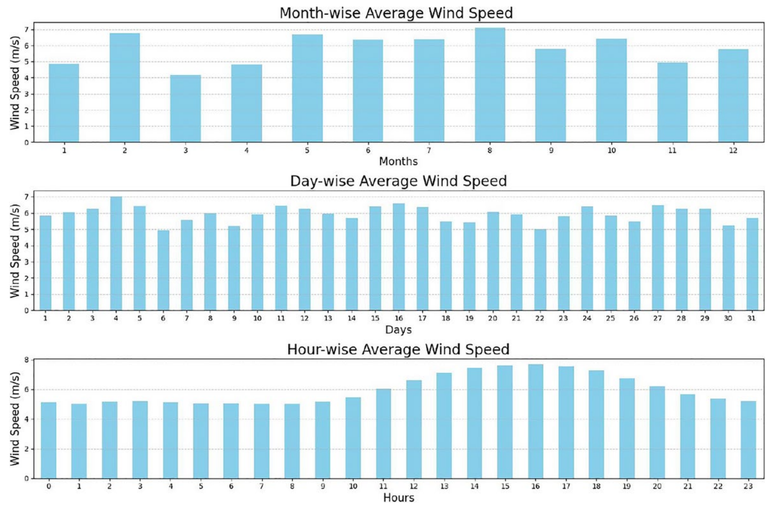

Fig. 1 Average wind speed throughout different range of time

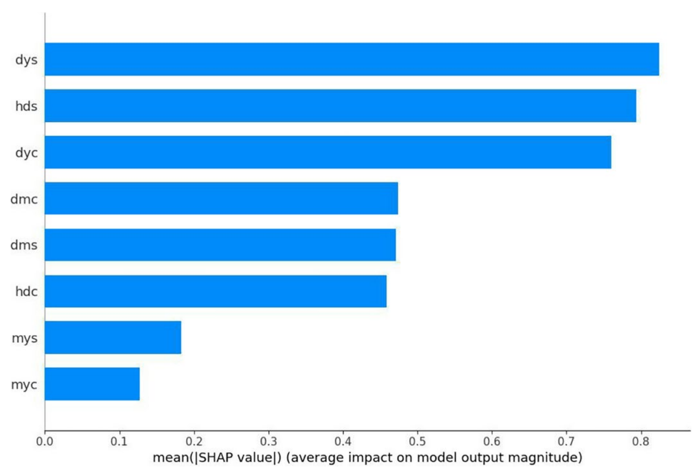

Fig. 2 Temporal feature importance on wind speed

Figure 1 consists of three bar charts, each depicting different time-based averages of wind speed. The top graph illustrates the monthly average wind speed, providing a broad overview of trends and seasonal variations throughout the year. The middle graph presents the daily average wind speed, offering insights into day-to-day fluctuations and patterns. Finally, the bottom graph shows the hourly average wind speed, highlighting the intra-day variations and capturing the finer details of wind speed across different hours. Together, these graphs display a comprehensive view of wind speed patterns across various time scales.

Temporal behaviour display multiple inherent cycles. Therefore, breaking down the timestamp into time-related features captures temporal patterns crucial for wind power prediction. Figure 2 shows how each piece of information in a forecasting model contributes to individual predictions. Cyclical transformations of time features effectively represent periodic behavior, helping machine learning models better understand inherent cycles, improving accuracy, and avoiding misinterpretation of time relationships. The day of the year, and the hour of the day mainly contributes for prediction. Moreover

, Power curtailment by wind power producers impacts grid power demand and ensures safe operations during harsh environmental conditions.

Overall, three types of information, temporal information obtained from the timestamp, metrological information obtained from weather data (wind speed, wind direction, temperature, humidity, pressure), and power curtailment information obtained from turbine signal data (blade pitch angle, rotor speed, relative angle to the wind direction) are causal variables for wind power production. We aim to identify an effective ML-based forecasting model by incorporating these variables, as shown in Fig. 3. In order to highlight the relationships between multiple time series variable, we define the problem of forecasting using multivariate time-series input, $X = \{x_{it}\} \in \mathbb{R}^{N\times T}$, where N is the number of variables and T is the number of time steps. The observed values at time step t are denoted as $X_t \in \mathbb{R}^N$. Given the observed values from the previous K time steps $X_{t-K}, \ldots, X_{t-1}$, the goal of multivariate time-series forecasting is to predict the next H time steps in as $\hat{X}_t, \hat{X}_{t+1}, \ldots, \hat{X}_{t+H-1}$. These future values are estimated by the forecasting model F which can either be provided as prior knowledge or learned from the data.

The model can be expressed as:

$$\hat{X}_t, \hat{X}_{t+1}, \ldots, \hat{X}_{t+H-1} = F(X_{t-K}, \ldots, X_{t-1})$$

Methodology

In this section, we outline the process of wind power forecasting, as illustrated in Fig. 4. Wind turbine power forecasting relies on historical power generation and numerical wind prediction data, with ML models for training. However, training efficiency decreases with large input data volumes. To ensure data quality, preprocessing steps are employed to remove duplicates, handle missing values, and eliminate outliers and anomalies. A SHapley Additive exPlanations (SHAP)-based feature selection method is applied to identify the most relevant features for the forecasting model. For each model evaluation, 80% of the data is used for training, and 20% is reserved for testing. Hyperparameter tuning is applied to both traditional machine learning and deep learning models to identify the optimal forecasting model. Finally, the performance of various models is analyzed and discussed to assess their effectiveness in wind turbine power forecasting.

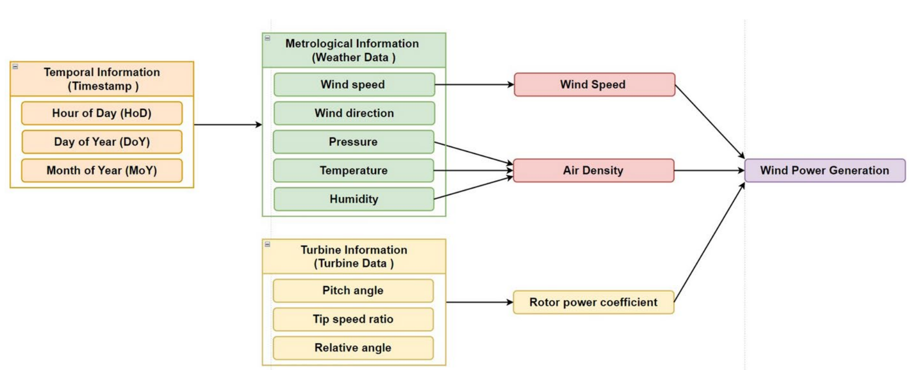

Fig. 3 Causal relationship between temporal information, metrological information and turbine signal information for wind power generation

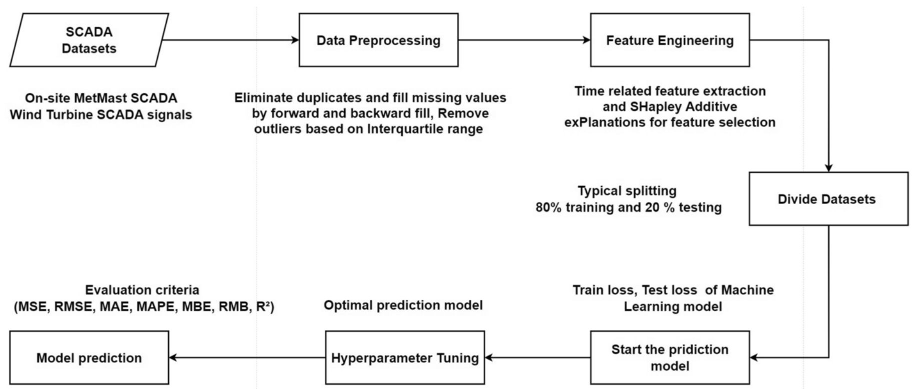

Fig. 4 Schematic diagram of the various steps involved in the power forecasting process implemented in this work

Table 2 Wind turbine operating conditions

Operating condition	Wind speed	Rotor speed	Power generation
Idel	In-sufficient	Stationary	Zero
Motoring	In-sufficient	Rotating	Negative
Shut down	Sufficient	Stationary	Zero
Free wheeling	Sufficient	Rotating	Zero
Normal operating	Sufficient	Rotating	Positive

Data preprocessing

To clean the tabular data, duplication is removed, and missing values are addressed using simple imputation techniques such as forward and backward filling, as described in [27]. Additionally, the interquartile range method is applied to detect and remove outliers in the measured signals, enhancing forecasting accuracy. Additionally, outliers are removed using the theoretical power curve of the wind turbine, supported by explainable reasons for motoring, shut down, and free wheeling. Wind turbines operate under five distinct conditions: idle, motoring, shutdown, free wheeling, and normal operating as shown in Table 2. In idle conditions, wind speed is insufficient, typically below the turbine's cut-in speed, causing the rotor blades to remain stationary and resulting in no power generation. During motoring, although wind speeds are inadequate for power production, the rotor continues to revolve, leading to negative power generation. The shutdown condition occurs when wind speed is adequate for power production, but the rotor is stationary, so no power is generated. In the free wheeling condition, the wind speed is sufficient, and the rotor is revolving, yet no power is being generated. Lastly, in normal operating mode, wind speed is adequate, the rotor is turning, and positive power generation is occurring.

Feature engineering

In this work, all three causal variables, the temporal variable, meteorological variable and turbine operating variable, are used for wind power forecasting.

Temporal variables

Four temporal features-hour of the day, day of the month, month of the year, and day of the year-are extracted and transformed into a cyclic representation to ensure continuity, enhance smooth transitions, and improve model interpretability and performance. Sine and cosine are similar, with a 90-degree phase difference between them, but using only one does not fully capture the position within a cycle. For example, $\sin(0)$ (the start of the cycle) and $\sin(\pi)$ (the middle of the cycle) both equal zero, making it impossible to distinguish between these points using sine alone. Therefore, both sine and cosine are used together to capture different aspects of the cyclic feature-both the position and the phase. Based on the SHAP values obtained from RF regression, as shown in Fig. 2, two temporal and four causal variables are selected for the forecasting model.

Meteorological variables

These consist of wind speed, wind direction, and air density. **Wind speed** is the primary factor in wind turbine power generation because the amount of energy produced is directly related to the wind's velocity. Higher wind speeds result in significantly more power output. **Wind direction** is crucial for wind turbine power generation because turbines need to face the wind to capture maximum energy. Proper alignment with the wind direction ensures optimal performance and efficiency. **Air density** is another dependent variable in wind power forecasting because it directly influences the amount of energy a wind turbine can extract from the wind. Wind power is proportional to air density, as it affects the mass of air flowing through the turbine. Higher air density increases the wind's kinetic energy, leading to more power generation, while lower air density decreases it. Since air density is influenced by altitude, temperature, humidity, and pressure, it must be considered for accurate wind power forecasting. Available weather data, namely **ambient temperature**, **humidity**, and **atmospheric pressure**, are utilized.

Power curtailment information

Wind power forecasting relies on the power curtailment variable, making relative wind direction, pitch angle, and tip speed ratio critical factors. **Pitch angle:** Blade pitch angle is crucial for wind turbine power generation as it controls the angle of the blades relative to the wind. Adjusting the pitch angle optimizes the turbine's ability to capture wind energy efficiently and maintain safe operation under varying wind conditions. The tip speed ratio, which connects wind speed to rotor RPM, is unavailable in this dataset; therefore, we use **rotor RPM** for power forecasting instead of tip speed ratio. The complete list of variables that have been used in forecasting are given in Table 3.

Table 3 Variables used for forecasting the power production from wind turbine

Variable	SCADA Name	Description	Features	Units
Timeline	Timestamp	Time resolution	Day of Year	
			Hour of Day	
Wind direction	Amb_WindDir_Relative_Avg	Average wind direction	Wind Direction	°
Wind speed	Amb_WindSpeed_Avg	Average wind speed	Wind Speed	m/s
Temperature	Avg_AmbientTemp	Average temperature	Temperature	$^\circ$C
Humidity	Avg_Humidity	Average humidity	Humidity	%
Pressure	Avg_Pressure	Average pressure	Pressure	millibar
Pitch angle	Blds_PitchAngle_Avg	Average blade pitch angle	Pitch angle	°
Rotor speed	Rtr_RPM_Avg	Average rotor rpm	Rotor speed	rpm
Power	Grd_Prod_Pwr_Avg	Average power supplied on grid	Power	kW

Forecasting models

There is not a clear distinction between interpretable and non-interpretable ML models; however, they can be classified based on the reasoning behind their forecasting, as shown in Fig. 5. Interpretable ML models explain how forecasts are made, allowing humans to understand the reasoning behind decisions [28]. In contrast, non-interpretable models remain difficult for humans to interpret and further categorized into explainable models and black-box models. Explainable ML clarifies why a model made a particular prediction, focusing on making the model's outputs understandable to humans. GNNs are one of the examples of an explainable ML model and operate within the graph domain to address the limitations of black-box models commonly found in many deep learning methods [29]. Deep learning-based neural networks have a layered structure and provide forecasting results without offering a clear, understandable rationale for how those outputs are generated [5, 17, 24]. The following subsection provides a description of each model used in this work.

Interpretable ML models

An interpretable ML model uses algorithms to learn patterns from labeled data, enabling predictions on new, unseen data. In this work, the model transforms historical time-ordered data into a supervised learning problem and capture complex patterns in the data for prediction.

LR is a basic machine learning technique used to model the relationship between a dependent variables and independent variables [30]. It assumes a linear relationship, represented by a straight line in simple linear regression. The model aims to find the best-fitting line or plane that minimizes the sum of the squared differences between the observed values and the predicted values. Because of its simplicity, interpretability, and effectiveness in capturing linear trends in data, it is widely used for power prediction and forecasting. The multivariate linear regression model can be expressed compactly in matrix form as shown in Eq.(4),

$$\mathrm{y} = \mathrm{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{4}$$

Where: y is an $n \times 1$ vector of observed dependent variable values, X is an $n \times (p+1)$ matrix of independent variables (including a column of ones for the intercept), $\boldsymbol{\beta}$ is a $(p+1) \times 1$ vector of coefficients, $\boldsymbol{\epsilon}$ is an $n \times 1$ vector of error terms. The coefficients β are estimated using the ordinary least squares method, which minimizes the sum of squared errors.

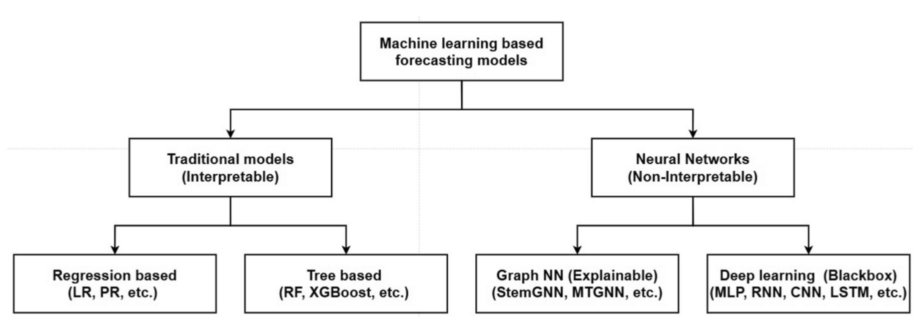

Fig. 5 Taxonomy of machine learning based forecasting model

KNN is a straightforward and intuitive algorithm for regression tasks, offering flexibility and adaptability at the cost of computational efficiency and sensitivity to noise [31]. Given a dataset with inputs $X = \{x_1, x_2, \ldots, x_n\}$ and corresponding outputs $Y = \{y_1, y_2, \ldots, y_n\}$, the prediction for a query point x_q is calculated as in Eq.(5),

$$\hat{y_q} = \frac{1}{k} \sum_{i \in N_k(x_q)} y_i \tag{5}$$

Where $N_k(x_q)$ denotes the indices of the *k* nearest neighbors of x_q, determined by a distance metric $d(x_q, x_i)$. Its effectiveness often depends on careful tuning of parameters like K and distance metrics, as well as preprocessing of data.

RF is an ensemble machine learning technique that improves prediction accuracy by combining multiple decision trees [32]. This method constructs a "forest" of decision trees during training and makes predictions by aggregating the outputs of these trees. For multivariate regression, the model predicts multiple dependent variables simultaneously. The prediction for a single output variable is shown as in Eq.(6),

$$\hat{y} = \frac{1}{M} \sum_{m=1}^{M} f_m(\mathrm{x}) \tag{6}$$

Where: $f_m(\mathrm{x})$ is the prediction of the *m*-th tree, *M* is the total number of trees, x is the input vector.

XGBoost was developed by Chen and Guestrin, is a highly efficient and scalable ML algorithm known for its exceptional performance [33]. XGBoost is based on the gradient boosting framework, which works by sequentially adding decision trees to improve predictions. Gradient boosting combines multiple weak learners (decision trees) into a strong one. It reduces bias and variance by iteratively correcting errors from previous trees. This can lead to better generalization in wind power forecasting, where the dynamics of the system may change over time. Each tree is built to correct the errors of the previous ones, refining the model's accuracy with each iteration. The overall objective function is minimized to optimize the trees and the model as shown in Eqs.(7),

$$\mathcal{O}(\theta) = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{7}$$

where: $\ell(y_i, \hat{y}_i)$: Loss function measuring prediction error (e.g., squared error for regression). - $\Omega(f_k)$: Regularization term penalizing model complexity to prevent overfitting. The final prediction for an input x_i is computed after all *K* trees have been built, summing their outputs to produce the model's prediction.

Explainable ML models

Explainable ML models are designed to make the prediction processes more interpretable to human cognition. Feature attribution methods like SHAP, attention mechanisms, subgraph extraction, and node and edge importance analysis are standard methods for enhancing interpretability in explainable models. GNN model can be made more interpretable by providing local neighborhoods within graphs [29]. A GNN framework for multivariate time series data that learns directed variable relations through a graph learning module, integrating external knowledge like variable attributes, was presented in [34]. In GNN-based forecasting, the graph structure is first discovered using causal discovery algorithms. Subsequently, the dependent values are estimated based on the inferred causal graph. There are numerous open-source Python packages available for initiating causal discovery [35, 36]. A typical GNN model combines several key modules: the Propagation Module for aggregating feature and topological information through convolution and skip connections to prevent over-smoothing, the Sampling Module for efficient propagation on large graphs, and the Pooling Module to extract high-level subgraph or graph representations, with these layers often stacked to enhance representational quality. In **Spectral Temporal Graph Neural Network (StemGNN)**, a graph was constructed to capture relationships among dependent variables, while spectral transformation embeds temporal seasonality [37]. This approach allows StemGNN to account for both inter-variable correlations and temporal dependencies, optimizing its forecasting capabilities.

Blackbox ML models

A black-box model has internal workings that are difficult to interpret, making its decision-making process challenging to understand due to the complexity and opacity of its underlying layers and parameters. **MLP** is a basic neural network with one hidden layer containing a reasonable number of neurons [38]. Overfitting the training data often leads to excellent performance on the training dataset but poor performance on the testing dataset. Dropout is a simple and effective method to improve the generalization ability of the neural network, especially when dealing with a small training dataset. **RNN :** An RNN features feedback connections that allow information to loop back from hidden to preceding layers, making it adept at processing sequential data [39]. By maintaining an internal state, RNNs can capture and leverage temporal dependencies in time series data, making them well-suited for tasks involving dynamic or ordered inputs. Typical RNNs often struggle with capturing long-term dependencies in data due to the vanishing gradient problem, where gradients used during back-propagation can become very small, hindering the learning of long-term patterns. LSTM networks address this issue by using specialized gating mechanisms that regulate the flow of information. **LSTM** networks are a type of RNN designed to excel at capturing long-term dependencies in sequential data [40]. They incorporate memory cells and gating mechanisms-input, output, and forget gates-that regulate the flow of information, allowing the network to retain relevant information over extended periods and mitigate the vanishing gradient problem. **GRU** is a simplified version of LSTM networks by using fewer gates-specifically, update and reset gates-while still managing to control the flow of information and mitigate the vanishing gradient problem [41]. GRUs are particularly effective in capturing temporal patterns and are often used in tasks such as time series forecasting due to their ability to remember and process information over varying time scales. The tuning parameters of various forecasting models used in this work are provided in Appendix A.

Data description and evaluation metrics

This section provides an overview of our analysis's data and evaluation metrics. The subsequent subsections detail the characteristics of the Supervisory Control and Data Acquisition (SCADA) data, followed by a description of the evaluation metrics

Year	Turbine signal (Timestamp)	Metrological (data)
2016	Turbine_ 01 (52439) Turbine_ 06 (50575) Turbine_ 07 (52445) Turbine_ 11 (52446)	(52697)
2017	Turbine_ 01 (52244) Turbine_ 06 (52346) Turbine_ 07 (52294) Turbine_ 11 (52352)	(34831)

Fig. 6 Wind turbine data description [42]

Fig. 7 Location of the wind turbine with respect to the origin reference point

employed, which offer a comprehensive assessment of model performance in power forecasting.

Data description

A SCADA system has been used to predict wind turbine power generation, and the data were collected over two years (2016 and 2017) at 10-minute intervals [42]. These publicly available datasets support condition monitoring research for wind turbines. The datasets, accessed on June 4, 2024, consist of five annual files: wind farm location, turbine log, failure log, turbine signal, and meteorological data. Both year's signal data contains the four turbines signal measurements with IDs T01, T06, T07, and T11. The timestamps are formatted as (2016-01-01 00:00:00+00:00) and for example, a forecasting horizon of 24 generates four-hour forecast values. The turbine signal consists of 83 different parameters, including wind speed, wind direction, blade pitch angle, rotor speed, and power production, and the meteorological file consists of 41 different parameters, including ambient temperature, humidity, atmospheric pressure as shown in Fig. 6. The relative location of the wind turbine is plotted with a reference in geographic coordinate systems in Fig. 7. In the 2017 dataset, meteorological variables for the first five months were missing, so those timestamps were excluded. The remaining data was merged into a single chunk of 87528 samples and then split, with 80% used for training and 20% for testing. Therefore, the entire dataset from 2016, along with a portion of the 2017 data, is primarily used for training, while the latter part of the 2017 data is reserved for testing.

For medium-term forecasting (one day), the data are resampled from 10-minute intervals to one-hour intervals, reducing the data size to one-sixth of the original.

Evaluation metrics

To evaluate the performance of deterministic forecasting models, the following metrics are used: mean square error (MSE), root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), mean bias error (MBE), relative mean bias error (RMBE), and coefficient of determination (R^2). MSE and RMSE measure forecasting accuracy, with RMSE providing a more interpretable metric by expressing error in the same units as power production. MAE provides a straightforward measure of average error magnitude, and MAPE offers a relative error measure as a percentage of the actual values. MBE provides insight into a model's bias by showing the distribution of its estimates and whether it systematically overestimates or underestimates the actual values. RMBE expresses MBE as a percentage of the mean of the actual values, allowing for a relative understanding of bias rather than an absolute one. A positive RMBE indicates that the model tends to overestimate the actual values, while a negative RMBE suggests that it underestimates them. The units of MAE, MBE, and RMSE match the units in the original measurements. R^2 value quantifies the proportion of variance in the predicted power, indicating the model's overall fit and explanatory power. The quantitative formulas of these metrics are shown in Eqs. (8-14), where N represents the number of samples, $Actual_i$ denotes the observed wind data, $Predict_i$ is the predicted wind power, and $\overline{\mathrm{Actual}}$ is the average of the actual power.

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(Actual_i - Predict_i)^2 \tag{8}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Actual_i - Predict_i)^2} \tag{9}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|Actual_i - Predict_i| \tag{10}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{Actual_i - Predict_i}{Actual_i}\right| \cdot 100 \tag{11}$$

$$MBE = \frac{1}{N}\sum_{i=1}^{N}(Predict_i - Actual_i) \tag{12}$$

$$RMBE = \frac{\sum_{i=1}^{N}(Predict_i - Actual_i)}{\sum_{i=1}^{N}Actual_i} \cdot 100 \tag{13}$$

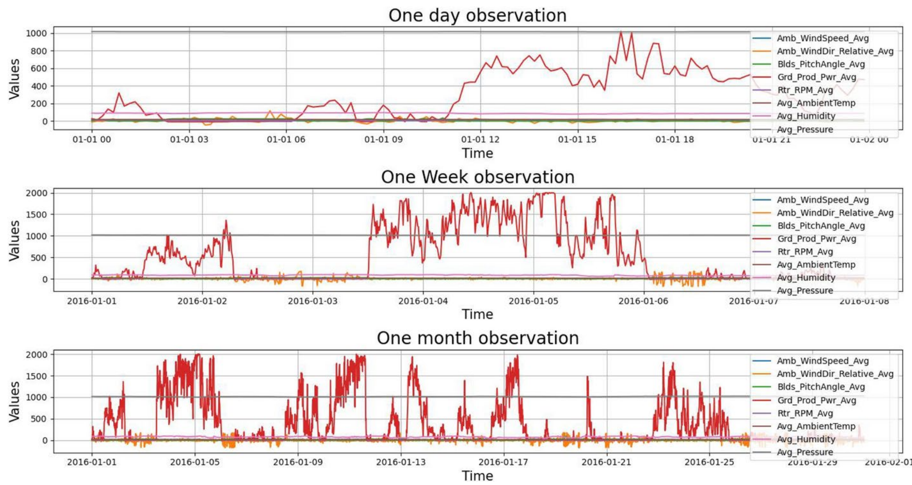

Fig. 8 Behavior of power forecasting variables at different time durations

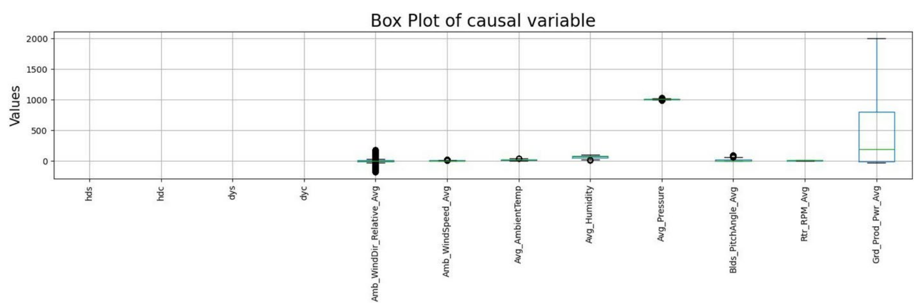

Fig. 9 Distribution of wind speed, pitch angle, rotor speed2, ambient temperature and humidity

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^{N} (Actual_i - Predict_i)^2}{\sum_{i=1}^{N} (Actual_i - \overline{Actual})^2} \tag{14}$$

Results

Figure 8 shows the three line plots, each depicting the behavior of weather and turbine variables influencing wind power production: wind speed, wind direction, pitch angle, rotor RPM, ambient temperature, humidity, and pressure. The top graph displays a 24-hour observation, capturing short-term fluctuations and immediate weather impacts on power generation. The middle graph provides a one-week observation, offering a broader perspective on trends and variability over several days. The bottom graph shows a one-month observation, revealing long-term patterns and seasonal effects. Together, these plots show a comprehensive view of how these environmental variables evolve over different time scales, highlighting their influence on wind power production.

The box plot displays the median (center line of the box), quartiles (edges of the box), and potential outliers (points outside the "whiskers"), offering a quick and clear understanding of the data's spread and symmetry. Moreover, the box plot demonstrates that the features are in different ranges. Figure 9 provides a concise summary of the distribution of dependent variables used in power forecasting. It illustrates the central tendency,

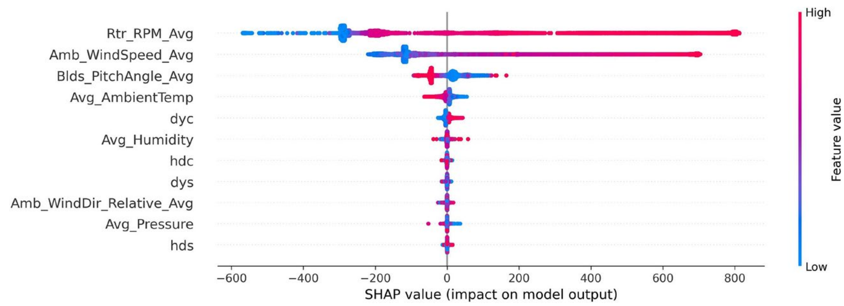

Fig. 10 Feature importance analysis for wind power forecasting using SHAP

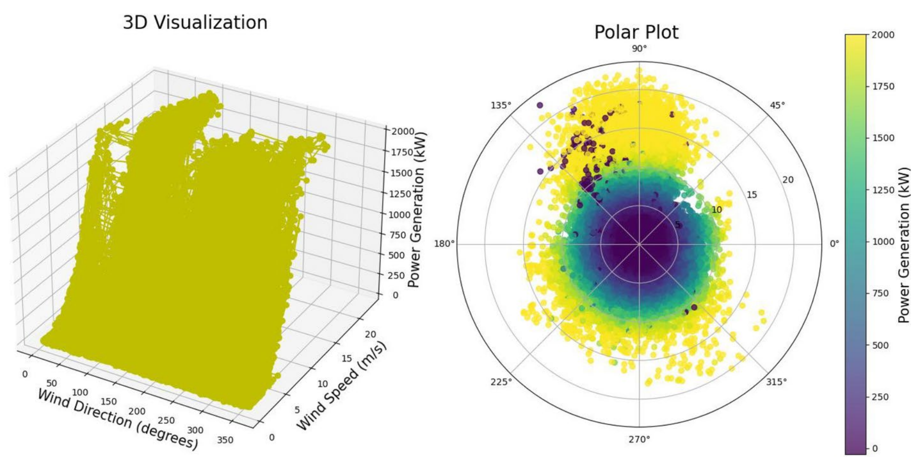

Fig. 11 3D plot and polar plot to show the relationship between wind direction, wind speed and power produced by wind turbine

variability, and overall distribution of the data. While some ML algorithms, such as RF and XGBoost, are not sensitive to the scale of features and thus do not require feature scaling, a min-max scaler is used for feature scaling in deep learning-based forecasting models.

Figure 10 shows the SHAP values explain how much different features contribute in wind power forecasting. Rotor speed is directly proportional to power production, making it the most significant factor, followed by wind speed, blade pitch angle, average temperature, cosine of the day of the year, humidity, and others. Wind speed and blade pitch angle influence the rotor power coefficient, highlighting their importance. Air density, which theoretically depends on ambient temperature, humidity, and pressure, also shows a small but reflected significance in the model.

Figure 11 shows the relationship between wind direction, wind speed, and power production in a wind turbine. The left graph presents a 3D plot illustrating how variations in wind speed and direction impact the turbine's power output. By plotting these three variables together, the figure highlights key trends, such as optimal wind conditions for maximum power generation and the influence of wind direction on turbine efficiency. This 3D visualization provides a comprehensive understanding of the complex interplay between these factors in wind energy production.

The right graph displays a polar plot that represents the relationship between wind direction, wind speed, and power production in a wind turbine. In this plot, wind direction is displayed around the circular axis, while the radial distance indicates wind speed.

The color intensity of the data points corresponds to the level of power production. This visualization effectively demonstrates how wind power varies with changes in both wind speed and direction, offering insights into the most productive wind conditions for the turbine. The graphs indicate that the wind direction around 80 to 120 degrees is particularly useful for normal power production.

The wind power versus wind speed scatter plot demonstrated a clear pattern that aligns with the turbine's power curve, offering valuable insights into the turbine's efficiency in converting wind energy into electricity across varying wind speeds, as shown in Fig. 12. This pattern not only highlights the predictable relationship between wind speed and power output but also serves as a crucial tool for evaluating turbine performance and optimizing energy production in real-world conditions. To remove outliers from a Turbine 1 with a total of 86,961 observation values, where specific conditions such as "Idle" (11,178 values highlighted by the red box), "Motoring" (14,504 values highlighted by the black box), "Shutdown" and "Freewheeling" (1,514 values by shutdown and 44 values by freewheeling highlighted by the purple box) are identified are abnormal operating conditions for power production. By removing these conditions, the graph on the right side shows the relationship between the wind speed (typically measured in meters per second, m/s) and the power output (kW) of a wind turbine 01. After removing those four types of operating conditions, some anomalous data values remain, likely due to power curtailment or sensor noise and degradation.

Wind turbine curtailment, the intentional reduction of power production despite sufficient wind, is typically implemented to maintain grid stability, manage oversupply, or address other operational constraints. This process is inherently probabilistic due to the uncertainty of future meteorological conditions and power demand [43]. Due to the lack of power demand data and grid signal details in the EDP, it is unclear how to explain the exact scenarios of the power curtailments highlighted by the yellow box in the left graph. To eliminate power curtailment anomalies, further investigation into the aerodynamic performance, specifically the relationship between wind pressure on the propeller and pitch angle, will be one strategy [44].

As demonstrated in Table 4 LR, KNN, and MLP, which rely solely on historical time-series patterns, demonstrate the lowest performance. The LR has the fastest forecast time (0.7189 sec) but higher errors (e.g., MAE: 119). Stem GNN, while requiring the most

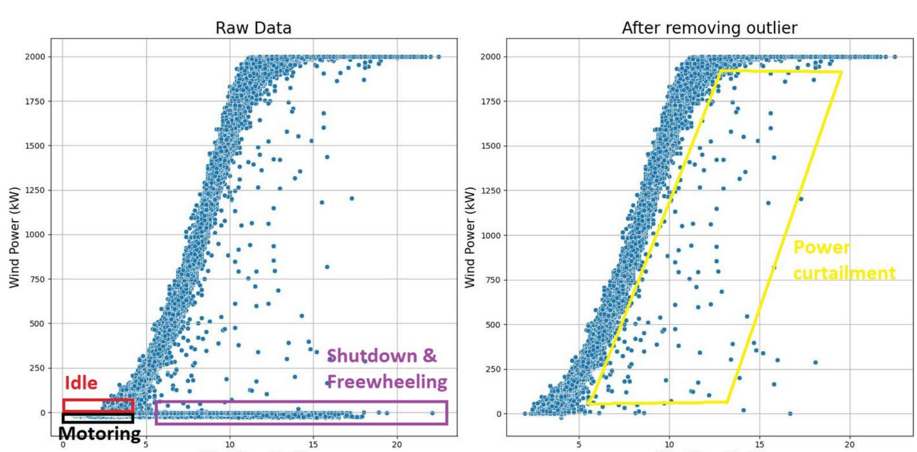

Fig. 12 Relationship between wind speed and power output

Table 4 Twenty minute forecasting performance of various machine learning models for wind turbine 1

Model	Forecast time (sec)	MAE	MAPE	MBE	RMBE	MSE	RMSE	R^2
LR	0.7189	119	5.65E+13	0	193	37468	193	0.8939
KNN	2.4981	147	2.67E+13	−10	242	59031	242	0.8328
XGBoost	2.9129	126	4.44E+13	−11	204	41722	204	0.8818
RF	114.8508	116	4.59E+13	0	194	38002	194	0.8924
Stem GNN	978.24	80	4561.53	4	134	18135	134	0.9335
MLP	47.5258	116	5.60E+13	9	195	38174	195	0.8919
RNN	242.9308	116	4.66E+13	−18	194	37867	194	0.8928
GRU	705.5143	118	4.59E+13	−27	195	38372	195	0.8913
LSTM	612.3831	117	5.43E+13	19	194	37726	194	0.8932

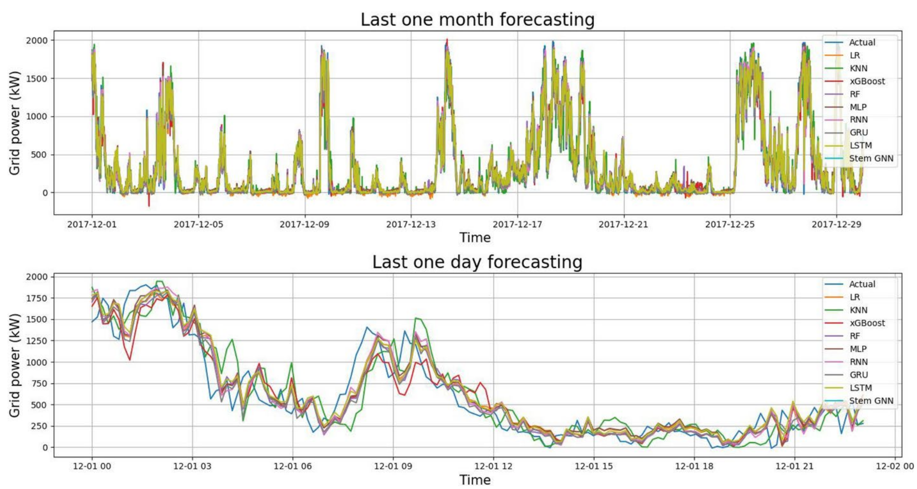

Fig. 13 Deterministic single-point wind power forecasting and the actual generated power from Turbine 01 using a 4-hour window with a 20 minutes horizon

significant computation time (978.24 sec), achieves the lowest MAE (80) and highest R^2 (0.9335), indicating superior accuracy. Models like RF, XGBoost, and MLP balance moderate computation times with competitive error metrics, while deep learning models such as RNN, GRU, and LSTM take longer but do not significantly outperform simpler models in terms of error and R^2 values. MAPE amplifies small differences in data, leading to disproportionately large scores from minor prediction errors in low-value data points. This can be misleading, especially in datasets with highly variable actual values, to show the actual prediction we have to look for Symmetric MAPE (SMAPE) of other metric like RMSE. Relying solely on numerical values can be misleading, as they may not fully capture the underlying patterns or variations in the data. Figure 13 provide clearer trends of forecasting by the StemGNN model. The top graph shows the one-month forecast, while the bottom graph presents an enlarged view of the one-day forecast values.

Figure 14 provides a comprehensive view of the strengths and weaknesses of various ML-based forecasting models with evaluation metrics stated in Section5.2. On the left side of the diagram, radar plots of different machine learning models (e.g., Linear Regression, KNN, Random Forests, XGBoost, StemGNN, MLP, RNN, GRU, and LSTM networks) and their forecasting performance are displayed. Parallel coordinate plots are an effective tool for comparing multiple variables and identifying patterns and relationships across different metrics, significantly when values overlap in radar plots. The

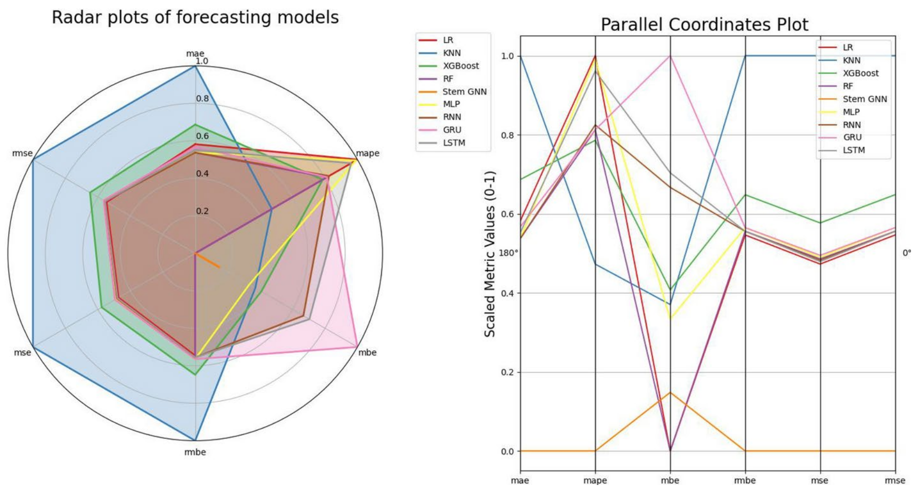

Fig. 14 Schematic diagram comparing the forecasting performance of various basic machine learning models on the left, and a comparison of these models using different evaluation criteria on the right

Table 5 Four hour forecasting performance of various machine learning models for wind turbine 01

Model	Forecast time (sec)	MAE	MAPE	MBE	RMBE	MSE	RMSE	R^2
LR	0.8268	302	1.24E+14	−4	416	173887	416	0.5076
KNN	2.1006	356	1.16E+13	−35	525	276270	525	0.2177
XGBoost	31.4843	314	3.06E+13	2	442	195987	442	0.445
RF	182.0792	289	5.67E+13	5	411	169429	411	0.5202
Stem GNN	7646.347	95	7482.23	17	150	22616	150	0.9173
MLP	77.1248	304	1.07E+14	6	433	188266	433	0.4669
RNN	948.4814	300	7.44E+13	−41	442	196153	442	0.4446
GRU	1460.9989	330	1.53E+13	38	489	239354	489	0.3223
LSTM	1545.3912	318	5.87E+12	15	479	229910	479	0.349

same results are shown on the right side, which clearly illustrate the evaluation metrics: MAE, MAPE, MBE, RMBE, MSE, and RMSE. By scaling the values from the minimum to 0 and the maximum to 1, StemGNN shows the best results across all metrics except MBE, which is 0.15 higher than LR and RF. Deep learning models struggle with forecasting despite the theoretical ability of neural networks with multiple hidden layers to fit any continuous function. They require more training time and optimization to improve performance. In contrast, both RF and XGBoost produce minor and similar prediction errors, but RF is more time-consuming due to the parallel construction of many independent decision trees. When comparing accuracy and computational time, XGBoost outperforms other methods, making it the better choice for this power forecasting task.

Tables 5 and 6 compare the performance of different machine learning models along with computation time and evaluation metrics stated in Section 5.2 for four-hour and one-day forecasting of wind turbine 1. Stem GNN shows the best accuracy for four-hour forecasting with the lowest MAE (95) and highest R^2 (0.9173), despite having a high computation time. Stem GNN again outperforms other models in one day forecasting, achieving a significantly lower MAE (230) and a higher R^2 (0.6557). In both scenarios, simpler models like LR and KNN have shorter forecast times but exhibit higher errors and lower R^2 values, indicating less reliable performance.

XGBoost provides better performance in forecasting results compared to other machine learning models in EDP, including MLP, RNN, LSTM, and GRU, due to

Table 6 One day forecasting performance for wind turbine 01 from one hour resampled observation

Model	Forecast time (sec)	MAE	MAPE	MBE	RMBE	MSE	RMSE	R^2
LR	0.1219	451	27.59	3	582	339362	582	0.0365
KNN	0.1704	518	31.18	23	687	473047	687	−0.343
XGBoost	19.7723	489	30.53	55	643	414537	643	−0.1769
RF	31.9214	466	29.15	25	599	359912	599	−0.0218
Stem GNN	1934.23	230	19.18	67	306	94188	306	0.6557
MLP	17.0502	488	32.56	67	621	386131	621	−0.0962
RNN	136.7218	454	26.19	−32	602	363081	602	−0.0308
GRU	202.9061	475	20.17	−119	657	432537	657	−0.228
LSTM	281.0324	503	27.51	−33	686	470626	686	−0.3361

structured tabular data because it builds decision trees that can capture complex interactions between features, outperforming neural networks, which are typically more suited for sequential or time-series data. Moreover, we have applied manual feature engineering, which transforms temporal information into sine and cosine at an hour of the day for short term dependencies and a day of the year for long-term dependencies. Neural networks typically rely on the raw sequential structure, which can be harder to optimize for forecasting. The weather pattern is similar but not identical to the replication, so temporal patterns in the weather variables are not very strong. As a result, RNN-based models like LSTM and GRU may struggle. However, XGBoost does not rely on time-series ordering to capture relationships between features, which explains why it performs better in our experiment.

We evaluate the computational time for each forecasting model. The results show that LR and KNN have the lowest training times, though at the cost of lower prediction accuracy. XGBoost, RF, and MLP require more computation time but provide accuracy comparable to the LR and KNN. Additionally, RNN, LSTM, and GRU take even longer than XGBoost and RF. Meanwhile, StemGNN demands significantly more training time and additional GPU resources. Regarding quick and fair forecasting, XGBoost demonstrates comparable performance across overall metrics. Since StemGNN effectively captures complex relationships between interconnected variables in a non-linearly, it outperforms other deep learning models in forecasting accuracy. Moreover, GNNs can capture both spatial and temporal information and provide dynamic representations of input features, where different variables may have varying importance at different times. This makes them the most accurate model among those presented in this work.

XGBoost is known for its fast training and tuning capabilities (e.g., via parallelization and optimized tree learning), which allows it to find optimal parameters quickly and efficiently. On the other hand, RNNs, LSTMs, and GRUs require more computational resources and longer training times, especially for long data sequences. Since dense network models are sensitive to data volume and quality, they typically need larger datasets to capture temporal dependencies well. XGBoost can perform better with smaller datasets and is less prone to overfitting due to its regularization techniques. Given the importance of quick forecasting, fair forecasting is preferred over delayed accuracy. Thus, the XGBoost model is recommended for overall performance. Table 7. presents a performance comparison of XGBoost using different input sequence lengths for wind power generation forecasting. Short-term forecasts are generally more accurate than long-term forecasts because of error accumulation, changing relevance of historical data, increased

Table 7 XGBoost-based forecasting for wind turbine power production with various window sizes and horizon

Window length	Horizon							
	One hour		Four hour		Eight hour		One day	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
One hour	218	325	333	467	393	529	510	669
Four hour	201	304	314	442	404	539	513	669
Eight hour	210	310	324	448	389	521	496	655
One day	201	301	309	441	382	527	500	653

uncertainty, and model limitations in capturing long-term dependencies. Therefore, as the forecast horizon increases, the accuracy tends to decrease. Therefore, selecting appropriate forecasting horizons at each update period is essential to create a complete time series, which can be achieved by minimizing the forecasting error.

Discussion

Wind power forecasting relies significantly on numerical wind data, which is critical for generating reliable predictions. To enhance the accuracy of forecasting model preformance, it is essential to consider various factors, including geographical location, which can affect wind patterns due to topography and land use. Additionally, understanding weather phenomena such as temperature fluctuations, pressure changes, and seasonal variations plays a crucial role in wind prediction. The complexities of atmospheric nonlinear processes also contribute to the intricacies of wind behavior. By integrating these diverse sources of information in models, the dynamic nature of wind patterns ultimately leads to improved wind power forecasting and better integration into energy systems. In summary, shorter forecasting horizons offer more accurate results but leave less time for wind power deployment, while longer horizons provide extended insights with reduced accuracy.

Resolution of historical window and horizon

Deterministic forecasting faces practical issues related to the historical window (the amount of past data used for forecasting) and the forecast horizon (the time period for which predictions are made). These issues can significantly affect the accuracy and reliability of forecasts.

The model may not capture important patterns, trends, or seasonality if the historical window is too short. This can lead to under-fitting, where the model misses key information and produces inaccurate forecasts. On the other hand, using too long a historical window can include outdated or irrelevant data, which can introduce noise and lead to over-fitting, Another limitation is that the deterministic models assume that future conditions will be similar to past ones. However, in many real-world scenarios, weather patterns evolve over time, and even a well-chosen window may struggle to adapt to significant changes. Similarly, longer forecast horizons increase uncertainty; the further out the prediction, the more likely it is that unforeseen variables will affect the outcome, leading to more significant forecast errors. Larger windows require more computational resources regarding processing power and memory, which can lead to practical constraints, especially in real-time forecasting scenarios.

Another key issue about forecasting is that it is highly sensitive to the quality of historical data. If there are gaps, errors, or biases in the past data, the model will propagate these issues into forecasted values. Moreover, deterministic forecasts do not provide information on the confidence of predictions. Deterministic forecasting typically lacks a feedback mechanism for improving the forecast as new data becomes available.

Extension

Integrating ML models with statistical methods and physics-based models, leveraging the strengths of both approaches, can further enhance their performance. Future research could develop hybrid forecasting frameworks combining ML techniques with physical and statistical models to improve forecasting accuracy and reliability.

An essential extension is the incorporation of probabilistic forecasting for power distribution planning. Unlike deterministic models, probabilistic forecasting provides a range of potential outcomes with associated probabilities, offering a more robust approach to managing the inherent uncertainties in real-world power planning, such as those related to weather, financial markets, and renewable energy production. Since risk analysis in condition monitoring is inherently probabilistic, understanding the likelihood of different outcomes for causal variables would enable organizations to make more informed, risk-aware decisions.

Additionally, the current analysis is limited by the availability of only turbine sensors and meteorological data, which is insufficient for comprehensive power curtailment analysis. Incorporating power grid observational data, along with load demand, and financial information could provide a more thorough understanding of wind turbine curtailment, expanding the industrial applications of wind power forecasting.

Conclusion

Accurate wind power forecasting is essential for energy planning, preventing imbalances, and meeting the renewable sector's demand for fast, reliable, and simple models. The nine ML-based data-driven forecasting models presented in this paper are highly significant for selecting the appropriate model for real-world applications. These models are validated using real operational datasets (turbine signal data and meteorological data), and their performance is benchmarked for comparison. Two different sampling intervals, ten minutes and one hour, are evaluated using various window sizes and forecast horizons, along with eight different evaluation metrics. StemGNN proves to be the most accurate model among those presented in this work regarding forecasting accuracy, except for computation time. XGBoost's superior handling of structured data, substantial feature utilization, efficiency in training, and robustness with smaller datasets often make it a better fit for tasks like wind power forecasting than neural network models that rely on sequential patterns.

Appendix A

For the Random Forest Regressor, the model consists of 50 decision trees, each with a maximum depth of 16. The K-Neighbors Regressor is configured with 3 neighbors for making predictions, and all neighbors' contributions are weighted equally. The deep learning-based prediction model architecture (MLP, RNN, GRU, and LSTM) includes three layers with dropout layers for regularization to prevent overfitting and a dense output layer to produce the final prediction. The model is compiled with the Adam optimizer and mean squared error loss function and then trained using 20 number of epochs and 64 batch sizes. The function returns the trained model and the training history. The StemGNN model was run along with GPU, utilizing 12.7 GB system RAM, 15.0 GB GPU RAM, and 112.6 GB disk space.

Abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Networks
ARMA	Auto Regressive Moving Average
CNN	Convolutional Neural Network
DT	Decision Trees
GNN	Graph Neural Network
GPR	Gaussian Process Regression
GRNN	Gated Recurrent Neural Network
KNN	K-nearest neighbor
LR	Linear Regression
LSTM	Long-Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
RF	Random Forests
RNN	Recurrent Neural Network
RMSE	Root Mean Square Error
SCADA	Supervisory Control and Data Acquisition
SHAP	SHapley Additive exPlanations
SMAPE	Symmetric MAPE
StemGNN	Spectral Temporal Graph Neural Network
SVM	Support Vector Machines
XGBoost	eXtreme Gradient Boosting

Acknowledgements
The authors thank Maneesh Singh, Knut Øvsthus, and Anne-Lena Kampen for their valuable discussions.

Funding
Open access funding provided by Western Norway University Of Applied Sciences.

Data Availability
The datasets used in this work are publicly available.

Declarations

Ethical Approval
Not applicable.

Received: 15 November 2024 / Accepted: 19 December 2024
Published online: 06 January 2025

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